

Xiangyu Zhang

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EDUCATION

Tsinghua University

B.S. in Computer Science and Technology

Sep. 2022 — Jul. 2026

GPA 3.8 / 4.0

University of California, Berkeley

Exchange Student, BGA Program

Jan. 2025 — May. 2025

GPA 4.0 / 4.0

University of California, Berkeley

PhD Student, IEOR

Aug. 2026 — Present

GPA - / 4.0

PUBLICATIONS

Heyang Yu*, Yinan Han*, **Xiangyu Zhang**, Baiqiao Yin, Bowen Chang, Xiangyu Han, Xinhao Liu, Jing Zhang, Marco Pavone, Chen Feng, Saining Xie, Yiming Li. **Thinking in 360°: Humanoid Visual Search in the Wild.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Weitao Li, **Xiangyu Zhang**, Kaiming Liu, Xuanyu Lei, Weizhi Ma, Yang Liu. **Efficient Dynamic Clustering-Based Document Compression for Retrieval-Augmented Generation.** *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)* 2025.

Lihang Pan*, Bowen Wang*, Chun Yu, Yuxuan Chen, **Xiangyu Zhang**, Yuanchun Shi. **AutoTask: Executing Arbitrary Voice Commands by Exploring and Learning from Mobile GUI.** *arXiv:2312.16062*

RESEARCH EXPERIENCE

Research Assistant

IEOR, University of California, Berkeley

Jan. 2025 — Present

Advised by **Prof. Zeyu Zheng**

- Challenge: Existing image-editing models rely on bounding boxes and masks to perform localized edits, limiting flexibility and requiring additional user input.
- Introduced a tuning-free, mask-free approach that leverages attention-map manipulation to achieve text-guided reference-object placement in arbitrary backgrounds.
- Constructed a 50k-scale synthetic dataset for object insertion and an additional 10k-scale variant with high-quality shadow removal.

Research Assistant

New York University

Sep. 2025 — May. 2026

Advised by **Prof. Saining Xie & Prof. Yiming Li**

- Introduced the first benchmark for embodied panoramic visual search and demonstrated that large multimodal models can be trained to perform human-like active head-rotation search in real-world environments.
- Contributed by annotating 1/3 of the dataset, conducting experiments, and iteratively refining RL training objectives.

Research Assistant

Yale University

May. 2025 — Apr. 2026

Advised by **Prof. Smita Krishnaswamy**

- Challenge: In the context of forecasting trajectories from irregularly sampled longitudinal medical images, the latent space lacks a clear separation between temporally relevant and irrelevant factors.
- Introduced a disentanglement loss to regularize the latent space, producing well-organized representations that better support trajectory prediction.

Research Assistant

Institute for AI Industry Research, Tsinghua University

Jun. 2024 — Apr. 2025

Advised by **Prof. Weizhi Ma**

- Introduced a retrieval-augmented generation framework that mitigates hallucinations and enhances answer accuracy through dynamic document clustering.
- Led a project that learns answer-style guidance from knowledge-irrelevant documents to strengthen question-answering.

Research Assistant

Tsinghua Pervasive Human Computer Interaction Lab

Aug. 2023 — Nov. 2023

Advised by **Prof. Yuanchun Shi**

- Introduced a new paradigm in which an agent performs arbitrary tasks in an OS environment.
- Presented a GUI-based agent workflow that allows users to automate any intent with a single command.

INDUSTRY EXPERIENCE

Chief Information Officer / Project Manager

Yixin Technology Inc.

Mar. 2024 — Dec. 2024

- Developed an application with mental health care and AI chatbot service.
- Led an IT team of ten students from UC Berkeley, University of Michigan, Tsinghua University, and etc.
- Led and participated in the full-stack implementation of back-end, front-end, UI design, deployment, and testing.

PERSONAL PROJECTS

Enhancing LLM's Coding Ability with Tree-Based Searching Methods

Project Link: https://github.com/Painkillerzzz/code_contest

Dec. 2024 — Jan. 2025

- Implemented and evaluated approaches, achieving notably higher pass rates and efficient use of computational budgets.
- Identified challenges with tree-based methods and proposed potential refinements to improve robustness and applicability in real-world scenarios.

I-Heart: A Chatbot for Mental Health and Psychotherapy

Mar. 2024 — Jul. 2024

- Developed a chatbot consists of fine-tuned LLMs and text-to-speech models.
- Provide clients with customized treatments, supported by professional conversation and a styled tune.

The Dance of Fire and Ice: A Hardware Music Game

Project Link: <https://github.com/Painkillerzzz/digital-logic-design>

May. 2024 — Jun. 2024

- Developed a music video game based on FPGA and hardware design language.

Generals: An AI Agent Competition Game

Project Link: <https://www.saiblo.net/game/35>

Sep. 2023 — Jan. 2024

- Developed an agent competition game and deployed it on a platform where competitors can submit their agents to fight against each other.
- Completed animations and model action control, participated in writing and debugging UI logic.

HONORS AND AWARDS

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| • Sports Excellence Scholarship | Tsinghua University, 2024 |
| • Literary and Art Excellence Scholarship | Tsinghua University, 2024 |
| • Social Practice Excellence Scholarship | Tsinghua University, 2023 |
| • Literary and Art Excellence Scholarship | Tsinghua University, 2023 |
| • Excellence Entrance Scholarship | Tsinghua University, 2022 |

SKILLS

- **Language:** English (TOEFL 102; GRE verbal 156, quantitative 170), Mandarin Chinese (native).
- **Programming:** Python, C/C++, Java, Kotlin, C#, System Verilog
- **Tool:** Pytorch, HuggingFace, Wandb, Unity3D